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Chapter 14

Respiration

Learning Objectives

The learner will be able to,

- *Recognize the stages of glucose breakdown and its redox system.*
- *Differentiate aerobic respiration from anaerobic respiration.*
- *Describe the conditions under which respiration occurs.*
- *Realize the role of mitochondria as power house of the cell.*
- *Understand, how ATP molecules are generated during respiration.*

Chapter Outline

14.1 Gaseous exchange
14.2 Structure of ATP
14.3 Redox reactions
14.4 Types of Respiration
14.5 Stages of Respiration
14.6 Respiratory Quotient
14.7 Anaerobic Respiration
14.8 Factors Affecting Respiration
14.9 Pentose Phosphate Pathway

CQ9UEZ

Plant and Animal Interdependence

In biosphere, plants and animals are complementary systems which are integrated to sustain life. In plants, oxygen enters through the stomata and it is transported to cells, where oxygen is utilized for energy production. Plants require carbon dioxide to survive, to produce carbohydrates and to release oxygen through photosynthesis. These oxygen molecules are inhaled by human through the nose, which reaches the lungs where oxygen is transported through the blood and it reaches cells. Cellular respiration takes place inside the cell. A specialized respiratory system is present in animals but is absent in plants for delivering oxygen inside the cell. But the cellular respiration stages are similar in both plants and animals which hint at evolutionary divergence.

